

Reliability Failure of Remaining-Useful-Life Prediction Intervals under Sensor Degradation: A Leakage-Audited Evaluation, Mean-Scale Attribution, and Maintenance-Decision Consequences

Yi Wang

Abstract—Prediction intervals on remaining useful life (RUL) feed maintenance decisions, so the reliability question is when they fail and what the failure costs. We evaluate five interval mechanisms (fixed variance, heteroscedastic likelihood, MC Dropout, deep ensemble, split conformal prediction) on two backbones and all four C-MAPSS sub-datasets under a leakage-audited protocol with engine-level training, validation and calibration sets. A controlled 2×2 retraining experiment, with the fitting engines held fixed, shows that test-set checkpoint selection moves RMSE by up to 14% and the NASA score by up to 106%, while the scope of the normaliser fit moves RMSE by at most 0.07 of a cycle. Under fair calibration, coverage rankings do not transfer between backbones; by interval score the deep ensemble is best in all eight backbone–dataset combinations in sample, while a paired comparison against independently seeded ensembles favours the ensemble in point estimate on all four sub-datasets and excludes zero on none. Under additive noise a four-way frozen-output decomposition attributes coverage loss predominantly to mean error in all 16 backbone–dataset–mechanism cases by point estimate, regardless of decomposition order, with a per-seed paired bootstrap excluding zero in 32 of the 32 order-specific cells. Structured degradation, applied to the sensor channels only, gives two separate findings: on single-condition data a within-window drift damages coverage at near-zero input excursion (below 10^{-4}), and on six-condition data drift and bias produce comparable excursion with very different consequences. Six controls support a monotone within-window trend as the damaging feature, strongly for the recurrent backbone and partially for the Transformer. In a terminal-instant decision stress test the errors are asymmetric: drift triggers premature maintenance on 52–69% of six-condition engines at a twenty-cycle lead time, with a mean true remaining life among those triggers of 101–105 cycles, whereas bias produces unrecognised maintenance needs, and cost rankings between mechanisms reverse across cost ratios in 36 of 96 conditions. A single excursion statistic does not characterise these risks.

Index Terms—Remaining useful life, prediction intervals, uncertainty quantification, sensor degradation, data leakage, conformal prediction, maintenance decisions, C-MAPSS.

I. INTRODUCTION

A PREDICTION interval on remaining useful life (RUL) is an engineering commitment: a maintenance planner who reads a 90% lower bound of twenty cycles will schedule

work on that basis. For the commitment to be reliable, three things must hold that are usually assessed separately or not at all. The interval must be calibrated on the data it was validated on; the validation must not have leaked information from the test set into the model; and the interval must degrade gracefully, or at least detectably, when the sensors that feed it degrade. This paper assesses all three on the NASA C-MAPSS benchmark [1], for five interval mechanisms and two backbone architectures, and then asks what the observed failures cost in maintenance terms.

The motivation is that the RUL literature has reached a state in which evaluation protocol matters as much as method. Point-prediction accuracy on C-MAPSS has plateaued, with new architectures separated by a few percent [2]–[5], and a growing number of methods report intervals [6]–[11]. Yet the protocol behind a C-MAPSS result is rarely stated at the level needed to reproduce it. A recent audit shows that window-level splitting alone can move accuracy on this benchmark from modest to near-perfect [12]. We show below that two further choices, how the checkpoint is selected and where the normaliser is fitted, move RMSE by more than the margins that distinguish published methods. Uncertainty comparisons inherit this problem and add one of their own: mechanisms that estimate their scale from training residuals are compared against mechanisms that use held-out calibration data, so that a difference in calibration resource is read as a difference in method.

Sensor degradation is the second reliability concern and the one this paper spends most of its effort on. Prognostic models are certified on healthy sensing chains and then operated, sometimes for years, on chains whose sensors drift, acquire offsets and lose gain. The reliability literature has long treated sensor and system degradation jointly [13], [14], but the uncertainty-quantification literature has evaluated intervals almost exclusively on clean inputs, and where robustness is studied it is under additive noise. We find that additive noise is the least informative of the degradations we test.

The paper makes four contributions.

- 1) A leakage-audited protocol for interval evaluation on C-MAPSS (Section III), with a controlled 2×2 retraining

experiment, the fitting engines held fixed, separating checkpoint selection from normaliser scope.

- 2) A fairly calibrated benchmark of five mechanisms on two backbones (Section VI), scored by proper interval scores rather than coverage alone, showing which rankings transfer between backbones and which do not.
- 3) An order-robust attribution of coverage loss under degradation to mean error versus learned scale (Section VII), using a four-way frozen-output decomposition.
- 4) An account of structured sensor degradation (Section VIII) in which within-window drift is shown, through six controls, to damage intervals through its temporal structure, and its maintenance consequence is quantified as premature rather than unrecognised maintenance (Section IX).

The paper proposes no new architecture. The audit of our own earlier pipeline, which motivated the protocol, is recorded in Appendix A rather than in the main text.

II. RELATED WORK

RUL prediction on C-MAPSS: Convolutional [15], [16], recurrent [17] and hybrid [2]–[4] architectures report FD001 RMSE of 11–13 cycles. Reviews [18]–[20] note that comparability is limited by undocumented preprocessing and split choices; Ruvaifa et al. [5] show a plain LSTM matching more complex models once preprocessing is controlled, and Shamim et al. [12] audit window-level leakage. Neither addresses checkpoint selection, normaliser scope, or interval metrics.

Uncertainty quantification in prognostics: MC Dropout [21], deep ensembles [22], heteroscedastic regression [23], [24] and evidential methods [9] have each been applied to RUL. Basora et al. [6] benchmark several on N-CMAPSS and find heteroscedastic networks competitive at low cost; our benchmark differs in its leakage audit, fair calibration, proper scoring and degradation study, not in the mechanisms compared. Conformal RUL prediction [7], [10], [11] rests on exchangeability of calibration and test scores [25]. Surveys [26], [27] document that predictive uncertainty degrades under shift, and Valdenegro-Toro and Saromo Mori [28] report in vision that aleatoric estimates are unreliable out of distribution while ensembles disentangle best; our attribution results are consistent with theirs. Seitzer et al. [29] and Stirn et al. [30] document pathologies of Gaussian likelihood training relevant to the variance-floor saturation we observe.

Measurement uncertainty and learned models: The GUM [31] presumes that the uncertainty of the measurement model is negligible relative to that of its inputs; recent metrological work asks when learned models satisfy this [32]. Section XI returns to this briefly.

III. EVALUATION PROTOCOL

A. Two Evaluation Units

C-MAPSS training files contain complete run-to-failure trajectories; test files contain trajectories truncated before failure with the terminal RUL supplied separately. Two evaluation units are used in this paper and are never mixed. The *terminal unit* is one prediction per test engine, from the last available

window, scored against the next-cycle target derived from the supplied terminal label (Section IV); all headline accuracy and interval figures (Tables I, II) use it. This differs from the usual current-cycle convention by one cycle in the linear region, so absolute accuracies are comparable with the literature only to that extent. The *trajectory unit* is every window of a test engine, with labels back-computed from the terminal label under the piecewise-linear target; it is used only for the per-engine attainment rate of Table II, where a single window per engine cannot define a per-engine coverage. Windows within an engine are dependent, and the trajectory-unit statistic is computed per engine first and then averaged with equal engine weight. The degradation study of Sections VII–IX uses the terminal unit throughout: each perturbed test engine contributes its last window, and coverage is averaged over engines, models and noise realisations. A provenance table listing, for every table in this paper, the generating script, label version, evaluation unit and averaging scheme is given in the supplementary material.

B. Engine-Level Three-Way Split with Disjoint Calibration

Each of the five seeds draws its own engine-level partition of the official training file of each sub-dataset, shared by every method for that seed: 60% of engines for fitting, 20% for validation (checkpoint selection only), and 20% for calibration (conformal quantiles and, under fair calibration, the residual scale of the fixed-variance and MC Dropout mechanisms). The official test file is touched once after all weights and quantiles are frozen. A run-time assertion aborts training if any engine appears in more than one subset. Splitting by engine rather than by window matters because a window-level split places overlapping segments of one trajectory on both sides [12]; keeping calibration apart from validation matters because a model selected on the calibration engines shrinks their residuals relative to unseen ones. The min–max normaliser is fitted per seed on the fitting engines only. This is stricter than fitting on the whole training file, and its cost is measured below rather than absorbed.

C. What the Two Most Consequential Protocol Choices Are Worth

Two decisions in a training loop can be made with or without reference to held-out data: which checkpoint to keep, and on which data to fit the normaliser. We trained the LSTM of Section IV with the same five seeds under all four combinations of checkpoint selection (on the official test set versus on the validation engines) and normaliser scope (the whole training file versus the fitting engines only). Table I reports the terminal-unit result for the three sub-datasets present in the earlier version of this work.

Holding the fitting engines fixed separates the two choices, and they are not of comparable size. Selecting the checkpoint on the test set improves RMSE by -1.94 to -0.29 cycles and the NASA score by -860 to -8 units; restricting the normaliser to the fitting engines changes RMSE by -0.04 to 0.07 cycles, at most 0.07 of a cycle, and the score by -4 to 214. The interaction, -0.43 to 0.47 cycles and -700 to -21 score units, is

TABLE I

CONTROLLED 2×2 PROTOCOL EXPERIMENT, HETEROSCEDASTIC LSTM, TERMINAL UNIT, MEAN OF FIVE SEEDS. ALL FOUR CELLS ARE TRAINED ON THE SAME FITTING ENGINES (THE SEED'S 60% FITTING SET); ONLY THE CHECKPOINT CRITERION AND THE NORMALISER SCOPE DIFFER. SEL.: CHECKPOINT SELECTED ON TEST SET (T) OR VALIDATION ENGINES (V). NORM.: NORMALISER FITTED ON WHOLE TRAINING FILE (W) OR FITTING ENGINES ONLY (F). SEL. IS THE MEAN OVER NORMALISER LEVELS OF (T-V), NORM. THE MEAN OVER SELECTION LEVELS OF (W-F), AND INT. THE DIFFERENCE OF DIFFERENCES (T/W-T/F)-(V/W-V/F), ALL ON THE RMSE/SCORE SCALE; NEGATIVE MEANS THE CHOICE MAKES THE RESULT LOOK BETTER.

		T/W	V/W	V/F	T/F	Sel.	Norm.	Int.
FD001	RMSE	13.48	15.31	15.41	13.36	-1.94	+0.01	+0.22
	Score	268.4	558.4	552.2	283.1	-279.5	-4.2	-20.8
FD002	RMSE	16.14	16.20	16.47	15.94	-0.29	-0.04	+0.47
	Score	1342.3	1388.1	1319.5	1349.2	-8.1	+30.8	-75.4
FD004	RMSE	19.35	20.37	20.08	19.49	-0.80	+0.07	-0.43
	Score	2750.5	3960.7	3396.0	2886.3	-859.9	+214.4	-700.4

of the same order as the normaliser effect and larger than it on FD004. The only cell in which a leaked choice does not flatter the result is the FD002 score under fit-only normalisation. From the fully leaked to the fully clean configuration the combined effect reaches 14% in RMSE and 106% in the NASA score on FD001, larger than the margins by which recent methods are separated; on the six-condition sub-datasets it is a few percent.

An earlier, uncontrolled version of this experiment left the fitting set free, so that the whole-file cells were also fitted on more engines; the normaliser then appeared to cost as much as the selection choice. It does not. That apparent effect was the training-set size, which is why the controlled design is the one reported here, and why a reader comparing protocols across papers should check what else changes with them.

The interval metrics move differently. Test-set selection changes PICP by -0.08 , -0.00 , -0.04 on the three sub-datasets and raises ECE on all three, by 0.003 to 0.021 . The selection criterion is RMSE, a statistic of the mean alone; the variance head is a bystander to the search, and whatever the search does to calibration is a side effect. A leaked point-accuracy figure is optimistic; a leaked interval figure is merely unreliable.

Whether these two choices are the most common in the C-MAPSS literature we cannot say without an audit of that literature, which we have not performed; we claim only that they are consequential and rarely stated.

IV. INTERVAL MECHANISMS AND FAIR CALIBRATION

Given a window of $W = 30$ cycles of F inputs—14 sensor channels on the single-condition FD001 and FD003, and 15 sensor channels plus the three operating-setting channels ($F = 18$) on the six-condition FD002 and FD004—the models predict $y = \min(\text{RUL}_{t+1}, 125)$, the clipped RUL at the cycle following the last observation in the window. This one-step-ahead target is the one used for training, calibration and checkpoint selection, and test engines are scored against $\min(\text{RUL} - 1, 125)$ derived from the official terminal label, so that every stage uses the same target instant. Scoring the same predictions against the current-cycle label instead changes RMSE by $+0.04$ to $+0.31$ cycles and PICP by -0.010 to $+0.008$ across backbones and sub-datasets. Two backbones are

used. The LSTM has two layers of 64 units with dropout 0.2. The Transformer projects each sensor vector linearly into a model dimension of $d = 64$, adds a learned positional encoding, and applies $n = 2$ encoder layers with $h = 4$ heads; the encoder output is average-pooled over the window before the heads. Hyperparameters of both were fixed before the protocol was adopted and were not re-tuned, so that Section III-C isolates the protocol effect; absolute accuracies are not upper bounds and every comparison is within a backbone. Training uses Adam at 10^{-3} , batch 256 and 150 epochs, keeping the checkpoint with the lowest validation RMSE, without early stopping; seeds are $\{42, 2024, 7, 888, 123\}$.

Five mechanisms form intervals from the same backbone. *Fixed variance*: an MSE-trained head with one global σ . *Heteroscedastic*: two heads emitting $\hat{\mu}$ and $\log \hat{\sigma}$ (clamped to $[-3, 2]$), trained by Gaussian negative log-likelihood [23], one deterministic pass. *MC Dropout*: the MSE backbone with dropout active, $T = 50$ passes, and the full predictive variance $\sigma_{\text{res}}^2 + \frac{1}{T} \sum_t (\hat{\mu}_t - \hat{\mu})^2$ [24]. *Deep ensemble*: the five heteroscedastic seeds combined by moment matching [22], whose variance contains both the spread of member means and the mean of member variances, so that member-level variance-floor saturation propagates into it. *Split conformal*: the $\lceil (n+1)(1-\alpha) \rceil / n$ quantile of $|y - \hat{\mu}| / \hat{\sigma}$ on the calibration engines scales the heteroscedastic interval (CP-norm); CP-norm inherits its width modulation from that head.

Fair calibration. The fixed-variance and MC Dropout mechanisms need a residual scale σ_{res} . Estimating it from training residuals, the usual practice, gives these two mechanisms less calibration resource than conformal prediction and makes their intervals optimistic: on every one of the eight backbone-dataset combinations the calibration-set residual scale is larger, by 8–36% on average and up to $1.95\times$ for one seed. All main-table figures below therefore estimate σ_{res} on the calibration engines, the same data CP-norm uses; the training-residual figures are given in Appendix B.

Metrics. RMSE and NASA score [1] for the point estimate. For intervals at nominal level $1 - \alpha = 0.9$: PICP, MPIW (cycles), ECE over twenty nominal levels, the per-engine attainment rate (trajectory unit; fraction of engines whose own coverage reaches 0.9), and the interval (Winkler) score $\text{IS} = (u - l) + \frac{2}{\alpha}(l - y)_+ + \frac{2}{\alpha}(y - u)_+$ [33], which penalises

width and miscoverage jointly and is proper. A single-level weighted interval score was also computed and identifies the same top-ranked mechanism in every backbone–dataset combination; it is given in the supplementary material, since with one interval level it adds little. We treat PICP as a constraint and IS as the quality measure; coverage above nominal is not better, it is wider.

V. DEGRADATION PROTOCOL

A. Additive Noise and the Excursion Axis

Gaussian noise is injected into the raw readings of the sensor channels before normalisation, without retraining; on FD002 and FD004 the three operating-setting inputs are never perturbed. The noise is set at levels referenced in three ways: to the within-condition channel standard deviation (scheme A), to the pooled standard deviation (B), and to a fixed fraction of the channel’s range over the official training file (C), at $\{0.1, 0.5, 1, 2, 5\}\%$. The range in C is the training-data range, not an instrument’s full scale; the channel statistics of all three schemes are computed over the whole official training file rather than the fitting engines only, so that the perturbation scale does not depend on the partition seed. Nominal signal-to-noise ratio is not comparable across schemes: at the same nominal level B injects an order of magnitude more absolute noise than A on six-condition data. Results are therefore expressed on the measured fraction of normalised inputs outside $[-1, 1]$, written f_{oob} , computed over the perturbed sensor channels of the same terminal windows that are fed to the model and scored, with each model’s own normaliser, and averaged over the same five models and five realisations as the coverage; results are pooled across schemes on that axis. The fixed-variance and MC Dropout mechanisms use the calibration-set residual scale of Section IV in every degradation experiment, as in Table II. Each level is a full factorial of five models and five noise realisations; within a realisation every mechanism receives the identical perturbed set, so that ensemble members read one sensor stream and cannot average the perturbation away.

B. Structured Degradation and Its Controls

Three deterministic degradations are applied at the same levels, again to the sensor channels only: a per-channel *bias* with random sign; a within-window *drift*, a linear ramp from zero to the nominal level at the last step; and a *gain* error multiplying raw readings by $1 \pm k$. To test the interpretation of the drift result, five controls are run at 5% of range on FD001 and FD002 for both backbones: a reversed ramp (maximum at the first step, zero at the last); a shuffled ramp with the same values in random temporal order; a single-channel ramp on s_9 only; a bias of magnitude equal to the ramp’s end value; and drift generated once on each continuous test trajectory and then windowed, so that a physical instant receives the same perturbation in every window that contains it.

C. Four-Way Attribution

Coverage can be lost because $|y - \hat{\mu}|$ grows or because $\hat{\sigma}$ changes. Writing μ_0, σ_0 for clean outputs and μ_1, σ_1 for

perturbed ones, we evaluate coverage under all four combinations $C(\mu_i, \sigma_j)$ and report the mean effect and scale effect under both decomposition orders together with their interaction. Attribution is stated only where both orders agree.

D. Maintenance Decision Rule

A fixed rule converts each interval into a decision at the terminal unit: maintenance is triggered when a one-sided 95% lower bound falls at or below a lead time $L \in \{10, 20, 30\}$ cycles. An engine with true RUL $\leq L$ that was not triggered is an *unrecognised maintenance need*; an engine triggered with true RUL $> L + 20$ is a *premature trigger*, and its remaining life beyond L is the *excess over the lead time*. Cost is $c_{\text{late}} r_{\text{late}} + c_{\text{early}} r_{\text{early}}$, reported at cost ratios $c_{\text{late}}/c_{\text{early}} \in \{5, 20, 100\}$.

VI. RESULTS ON CLEAN INPUTS

Table II gives the fairly calibrated benchmark.

Three findings hold across the table. First, coverage and interval score disagree about MC Dropout, and coverage rankings among the single-model mechanisms do not transfer between backbones. Under fair calibration MC Dropout covers at least as well as the heteroscedastic head in all eight backbone–dataset combinations, yet its interval score is the worst or second-worst of the five on the LSTM (74.3 against 62.4 for the head on FD002) and only competitive on the Transformer: it buys coverage with width. Between the fixed-variance reference and the heteroscedastic head the sign of the coverage difference flips across sub-datasets in a pattern that differs between backbones (the head covers less on FD001 and FD003 for the LSTM but only on FD002 for the Transformer). Nor does the ranking by degradation tolerance, reported in Section VII. Second, the ranking by interval score is more stable than the ranking by coverage: the deep ensemble has the lowest IS in 8 of the eight backbone–dataset combinations and the fixed-variance reference the highest in 7, with the single-pass heteroscedastic head between them. This is an in-sample observation on one ensemble per combination, and the ensemble of Table II is a cross-partition ensemble: because the partition of Section III is drawn per seed, its five members were fitted on five different subsets whose union covers 99 of the 100 FD001 training engines, so part of its advantage could be data coverage rather than averaging. Table III removes that confound by training five initialisations on one partition. On this partition the same-partition ensemble beats the mean of its own members by 2.1–9.0 IS points in all eight combinations, and the two ensembles differ by at most 5.1 points with no consistent direction. This shows that the gain does not depend entirely on cross-partition data coverage; it does not estimate the coverage effect over the space of partitions, since one partition was examined. The same table also shows that on FD001 five single models on this partition average 60.2 IS against 67.7 for five single models on five partitions, a difference as large as the ensemble margin, which is one reason five seeds on five partitions are a coarse instrument. Three disjoint five-member ensembles from a fifteen-seed pool

TABLE II

CLEAN-INPUT BENCHMARK UNDER FAIR CALIBRATION (CALIBRATION-SET RESIDUAL SCALE FOR FIXED VARIANCE AND MC DROPOUT). TERMINAL UNIT EXCEPT PER-ENGINE (TRAJECTORY UNIT). MEAN OF FIVE SEEDS; ENSEMBLE IS A SINGLE $M = 5$ POINT ESTIMATE. LOWER IS IS BETTER.

	Mechanism	PICP	MPIW	LSTM			Transformer				
				ECE	Per-eng.	IS	PICP	MPIW	ECE	Per-eng.	IS
FD001	Fixed variance	0.870	47.82	0.062	0.55	70.8	0.876	45.26	0.079	0.61	67.9
	Heteroscedastic	0.866	42.20	0.035	0.69	67.7	0.888	43.40	0.041	0.77	65.9
	MC Dropout	0.890	50.31	0.072	0.62	69.2	0.914	48.85	0.093	0.72	62.9
	Deep ensemble	0.910	45.05	0.032	0.81	63.1	0.910	45.40	0.065	0.81	60.1
	Split-CP (norm)	0.880	45.60	0.038	0.75	67.2	0.896	43.85	0.056	0.75	64.7
FD002	Fixed variance	0.912	60.92	0.059	0.60	74.1	0.919	58.91	0.080	0.63	72.9
	Heteroscedastic	0.913	51.63	0.030	0.75	62.4	0.896	49.79	0.032	0.73	64.7
	MC Dropout	0.927	62.82	0.067	0.64	74.3	0.919	60.52	0.079	0.63	71.8
	Deep ensemble	0.964	54.63	0.058	0.81	58.6	0.933	52.52	0.046	0.80	60.6
	Split-CP (norm)	0.932	53.72	0.037	0.77	62.9	0.915	52.60	0.028	0.76	65.1
FD003	Fixed variance	0.876	46.80	0.104	0.70	72.4	0.842	42.71	0.090	0.67	70.5
	Heteroscedastic	0.860	37.08	0.042	0.72	62.7	0.880	39.14	0.048	0.76	63.4
	MC Dropout	0.890	49.15	0.114	0.72	71.0	0.888	45.29	0.088	0.72	62.4
	Deep ensemble	0.910	40.63	0.076	0.76	56.7	0.930	42.86	0.082	0.84	56.0
	Split-CP (norm)	0.898	46.05	0.073	0.78	62.2	0.884	44.76	0.085	0.83	65.0
FD004	Fixed variance	0.877	60.93	0.051	0.68	87.8	0.853	55.53	0.074	0.68	88.4
	Heteroscedastic	0.882	58.01	0.027	0.76	80.6	0.866	50.48	0.029	0.75	84.0
	MC Dropout	0.886	63.07	0.056	0.70	87.4	0.881	57.66	0.038	0.74	81.0
	Deep ensemble	0.932	61.47	0.042	0.81	74.8	0.907	55.06	0.055	0.82	75.9
	Split-CP (norm)	0.882	56.72	0.046	0.73	80.3	0.866	51.74	0.094	0.76	84.2

TABLE III

ENSEMBLE GAIN WITH THE PARTITION HELD FIXED. SINGLE MODELS: FIVE INITIALISATION SEEDS TRAINED ON THE SEED-42 PARTITION; SAME-PARTITION ENSEMBLE: THOSE FIVE COMBINED; CROSS-PARTITION ENSEMBLE: THE FIVE SEEDS OF TABLE II, EACH TRAINED ON ITS OWN PARTITION. COLUMN 3 IS THE TABLE II HETEROSCEDASTIC ROW FOR REFERENCE (FIVE SEEDS, FIVE PARTITIONS). SOURCE:

RESULTS/SEEDS/SAMESPLIT_ENSEMBLE_CONTROL.JSON.

Backbone	Data	IS				PICP	
		5 part., single	1 part., single	1 part., ens.	5 part., ens.	1 part., ens.	5 part., ens.
LSTM	FD001	67.7	60.2 $\pm$ 2.2	58.0	63.1	0.890	0.910
LSTM	FD002	62.4	63.5 $\pm$ 3.9	58.7	58.6	0.917	0.964
LSTM	FD003	62.7	61.2 $\pm$ 5.3	57.2	56.7	0.920	0.910
LSTM	FD004	80.6	81.7 $\pm$ 3.6	77.7	74.8	0.928	0.932
Transformer	FD001	65.9	61.8 $\pm$ 3.2	59.5	60.1	0.910	0.910
Transformer	FD002	64.7	67.1 $\pm$ 6.0	62.2	60.6	0.925	0.933
Transformer	FD003	63.4	64.3 $\pm$ 6.0	55.3	56.0	0.920	0.930
Transformer	FD004	84.0	83.4 $\pm$ 4.5	75.7	75.9	0.899	0.907

(Appendix D) give IS spreads of 2–5 points, and a per-engine paired bootstrap on the same test set, of one disjoint ensemble against the single seed-42 model, excludes zero on none of the four sub-datasets, although the point estimate favours the ensemble on all four; this is a within-benchmark check of one comparison, not an external validation, and the margin is consistent in direction but not established per sub-dataset. On the LSTM the head recovers about 58% of the ensemble’s IS improvement over the fixed-variance reference (41–76% across the four sub-datasets) at one fifth of the inference cost. Third, fair calibration matters: with a training-residual scale the fixed-variance and MC Dropout rows are 8–36% narrower and correspondingly under-cover, and the heteroscedastic head’s apparent advantage over them is larger than it is here.

Inference cost is reported in Appendix E as distributions over repeated measurements at batch sizes 1, 32 and 512, without a pass/fail criterion; the architectural multipliers $T = 50$

and $M = 5$ are consistent with the LSTM measurements and are used as a diagnostic only.

VII. ATTRIBUTION OF COVERAGE LOSS UNDER ADDITIVE NOISE

Under noise, 33 of the 40 (backbone, sub-dataset, mechanism) combinations lose ten points of coverage before f_{oob} reaches 1.7%, and all but one before 4.3%. The exception is the Transformer ensemble on FD001: on the original grid its crossover interpolates to 10.2%; on a refined grid between 1.0% and 11.8% excursion the grand-mean curve crosses at 3.7%, but the curve is not monotone, dipping below the threshold and recovering more than once, and the first crossing of the five individual realisations ranges from 0.9–5.1%. The crossover is ill-defined for this cell; its grand-mean value is comparable with the largest of the other cells (4.3% for the ensemble Transformer on FD003), and we report it as an unstable outlier rather than as a stable threshold. Each noisy

condition is one seeded realisation of five trials, and counts of this kind move by a few cells between independent seedings. The relative ordering of mechanisms is not stable: on the LSTM the fixed-variance and MC Dropout rows fail last on FD001 and FD002 and first on FD003 and FD004, and on the Transformer the ordering agrees with the LSTM on none of the four sub-datasets. We report no mechanism as more tolerant than another.

What is stable is the attribution. Table IV gives the four-way decomposition at the point where the heteroscedastic head has lost ten points of coverage, obtained by interpolation on the measured curves for both backbones; C_{11} follows from the definition of the point, C_{10} is interpolated on the frozen-scale curve, and C_{01} is interpolated between two new evaluations that bracket the point. Full four-way values and cross-seed standard deviations (0.02–0.08 in coverage) are in the supplementary material; the sign judgements are robust to that spread, the magnitudes less so. An earlier draft read the decomposition at the nearest measured grid point instead, where the total loss ranged from 0.04 to 0.15; moving to the interpolated point changed the sign of the Transformer’s scale effect in five of eight cells and removed the one LSTM case in which mean dominance had depended on decomposition order, which is a warning about how sensitive this quantity is to where it is read. In all 16 (backbone, sub-dataset, mechanism) cases the seed-pooled point estimate of the mean effect exceeds that of the scale effect in magnitude under both decomposition orders. A per-seed paired bootstrap of $|\Delta_\mu| - |\Delta_\sigma|$ at the same interpolated point excludes zero in 16 of 16 cells under the mean-first order and in 16 of 16 under the scale-first order. Mean dominance is therefore supported by interval as well as by point estimate in 32 of the 32 order-specific cells. The scale effect is secondary in magnitude but not stable in sign. Read with the mean varied first, the learned scale appears to help in 2 cells (LSTM/FD004/het., LSTM/FD004/CP); read with the scale varied first, it is harmful in all sixteen, and averaged over the two orders it is harmful in 16. The interaction between mean and scale, $(C_{11} - C_{10}) - (C_{01} - C_{00})$, is positive in 16 cells, reaching +0.043 of coverage, and is what makes the sign order-dependent. We therefore report mean dominance as the finding and draw no conclusion about whether the learned scale ever helps under degradation: there is no order-independent evidence that it does. Shares should be read to within several points, since they move between independent noise realisations. We therefore report mean dominance as the finding and draw no conclusion about whether the learned scale ever helps under degradation: there is no order-independent evidence that it does, and under one order it never does.

Where the scale is harmful its behaviour is the same: $\hat{\sigma}$ contracts while $|y - \hat{\mu}|$ grows, because a rising fraction of samples reaches the architectural floor on $\log \hat{\sigma}$; the mean does not collapse to a constant but disperses, so the network is extrapolating. A hypothesis that the scale’s contribution is predicted by the clean-data floor fraction ordered the first three sub-datasets perfectly and was tested by adding FD003, whose floor fraction is the highest of the four but whose scale

contribution is near zero; it is not supported.

VIII. STRUCTURED DEGRADATION

A. Bias and Gain Are Tolerated; Drift Is Not

Table V gives the excursion at which the heteroscedastic head loses ten points of coverage under each degradation, both backbones. Under bias and gain the six-condition sub-datasets tolerate 4–9% excursion before losing ten points, an order of magnitude beyond the median 1.0% at which noise causes the same loss, and on the single-condition sub-datasets the criterion is reached only sporadically, with the reaching cells changing between noise realisations. The dose–response curves of the four degradations do not coincide on f_{oob} : mean absolute differences in PICP at matched excursion are 0.08–0.35. The excursion at which the ten-point criterion is met is similar across mechanisms, within about two points in any (degradation, sub-dataset) cell, but the depth of the collapse at 5% of range is not: coverage differs between mechanisms by 5–40 points there, and in 21 of the 24 (backbone, degradation, sub-dataset) cells the worst mechanism is one of the two that use the heteroscedastic scale head (the head itself or CP-norm), while the fixed-scale mechanisms retain up to 25 points more coverage and the ensemble or MC Dropout is best in every cell. Under the structured perturbations of this study, the mechanisms with a fixed residual scale generally retain higher coverage; since the mechanisms also differ in mean prediction, training objective and interval width, this is not attributed to the fixed scale alone, and higher coverage here does not translate into a better interval score or a lower maintenance cost.

Drift is different. At 5% of range a within-window ramp reduces the heteroscedastic head’s coverage to 0.64–0.64 on FD001, 0.03–0.05 on FD002, 0.10–0.21 on FD003 and 0.05–0.31 on FD004 across the two backbones, while on the single-condition sub-datasets f_{oob} remains below 10^{-4} (zero to numerical precision on FD001): windows near end of life already sit inside the training range and a ramp does not push them out. On the six-condition sub-datasets f_{oob} under drift at 5% of range is 9–11%, comparable with the 13–16% of bias, yet the coverage loss is far larger, and the ten-point criterion is met at 2.2–5.4% excursion rather than at the 4–9% of bias and gain.

a) *Operating-setting inputs.*: On FD002 and FD004 the models also receive three operating-setting channels, which all perturbations above leave untouched. A diagnostic on the heteroscedastic head at 5% drift separates their role. Perturbing only the settings raises the premature-trigger rate to 0.12–0.23 on FD004, with coverage between 0.59–0.89 across the two six-condition sub-datasets; perturbing sensors and settings together raises it from 0.52–0.59 (sensors only) to 0.67–0.68 on FD004, while on FD002 the two are indistinguishable (0.69 against 0.70). The sensor-only figures reported throughout are therefore the conservative ones; corrupted operating-setting inputs add to the decision error on FD004, and would need to be considered separately in any deployment where they are themselves measured.

TABLE IV

FOUR-WAY ATTRIBUTION OF COVERAGE LOSS UNDER ADDITIVE NOISE AT THE TEN-POINT LOSS OF THE HETEROSCEDASTIC HEAD. SCALE CONTRIBUTION, AS A SHARE OF THE TOTAL TEN-POINT LOSS (POSITIVE: THE LIVE SCALE MAKES THE LOSS LARGER), IS GIVEN UNDER BOTH DECOMPOSITION ORDERS. THE POINT-ESTIMATE ROW INDICATES WHETHER $|\text{MEAN}| > |\text{SCALE}|$ UNDER BOTH ORDERS ON THE SEED-POOLED PREDICTIONS; THE CI ROWS REPORT, FOR THE HETEROSCEDASTIC HEAD AND CP-NORM RESPECTIVELY, WHETHER A PER-SEED PAIRED BOOTSTRAP OF $|\Delta_\mu| - |\Delta_\sigma|$ AT THE SAME INTERPOLATED POINT EXCLUDES ZERO, SEPARATELY FOR EACH ORDER. SCALE CONTRIBUTION IS THE SHARE OF TOTAL LOSS. LSTM SHOWN; TRANSFORMER IN APPENDIX C.

		FD001	FD002	FD003	FD004
Heteroscedastic	scale, mean first	+28.3%	+8.9%	+3.6%	−5.8%
	scale, scale first	+37.5%	+30.4%	+27.0%	+36.9%
CP-norm	scale, mean first	+28.4%	+14.3%	+9.8%	−6.4%
	scale, scale first	+35.6%	+28.8%	+28.1%	+33.6%
Mean dominates (point estimate), both orders		yes	yes	yes	yes
CI excludes zero, mean first (het./CP)		yes/yes	yes/yes	yes/yes	yes/yes
CI excludes zero, scale first (het./CP)		yes/yes	yes/yes	yes/yes	yes/yes
Baseline clamp fraction		0.34	0.22	0.40	0.076

TABLE V

EXCURSION f_{oob} AT WHICH THE HETEROSCEDASTIC HEAD HAS LOST TEN POINTS OF COVERAGE UNDER STRUCTURED DEGRADATION. N.R.: NOT REACHED WITHIN 5% OF RANGE. VALUES ABOVE 10% ARE OUTSIDE THE REGIME STUDIED AND ARE GIVEN AS > 10 . $< 10^{-4}$: COVERAGE FALLS BY TEN POINTS WHILE THE EXCURSION STAYS BELOW 10^{-4} AT EVERY LEVEL, SO NO CROSSOVER ON THE EXCURSION AXIS EXISTS.

Backbone	Degradation	FD001	FD002	FD003	FD004
LSTM	Bias	n.r.	9.47%	n.r.	8.24%
	Drift	0.00%	2.53%	$< 10^{-4}$	5.40%
	Gain	n.r.	6.35%	> 10	5.18%
Transformer	Bias	n.r.	8.30%	0.03%	6.70%
	Drift	0.00%	2.22%	$< 10^{-4}$	2.94%
	Gain	n.r.	4.29%	> 10	4.15%

B. Why Drift Is Damaging: Five Controls

Table VI reports the controls on FD002. A bias of magnitude equal to the ramp's end value leaves coverage at 0.49 (LSTM) and 0.23 (Transformer) where the ramp leaves it at 0.05 and 0.03, at comparable excursion; the end magnitude is not the cause. A reversed ramp, with the same values in the opposite temporal order, leaves coverage at 0.68 (LSTM) and 0.69 (Transformer); a shuffled ramp at 0.29 on the LSTM but only 0.12 on the Transformer, which is nearly as damaged by the shuffled values as by the ramp; a ramp on the single channel s_9 , with excursion 10^{-4} , still reduces coverage to 0.63 and 0.54. Continuous drift is defined as a ramp from zero at the first cycle of each test trajectory to k times the range at its last cycle, after which the terminal window is taken as usual; for a median FD002 trajectory of 132 cycles its local slope within the window is 4.5 times gentler than the per-window ramp's, and its window mean offset about 89% of the end value against 50%. It produces the same loss as the per-window ramp (0.04 against 0.05 on the LSTM, 0.02 against 0.03 on the Transformer), while the equal-end bias, with 100% offset and zero slope, does not. Neither slope steepness nor offset magnitude is the driver; the presence of monotone within-window change is. A further control fixes the last k steps of the ramp and shuffles the rest: for the LSTM, $k = 1$ leaves coverage at 0.29, close to the fully shuffled 0.29, while $k = 5$ brings it to 0.13, approaching the ramp's 0.05; the damage requires a run of correctly ordered steps at the end of the window, not merely the end value. For

the Transformer every shuffled variant is nearly as damaging as the ramp (0.03–0.07), so the control does not discriminate there. Together the controls identify a monotone trend rising towards the prediction instant as the damaging feature for the recurrent backbone; for the Transformer the reversed-ramp and equal-bias controls point the same way but the temporal-order component is not established. Whether the models are reading the trend as degradation, or responding to it through some other nonlinearity, the present controls cannot say. A monotone trend towards the prediction instant is also the signature of degradation in the training data, so one reading is that a drifting sensor presents a healthy engine as one approaching failure; the present controls cannot establish that reading, only that the trend is what damages. What is observed is that the learned scale contracts under drift, as it does near end of life on clean data, and that the heteroscedastic head's coverage under drift is below the fixed-variance reference on three of four sub-datasets on both backbones. We note that the reversed and shuffled controls also reduce the perturbation at the final step, so they are consistent with recency weighting as well as with trend reading; the equal-magnitude bias control is the one that separates the two, and it favours trend.

IX. MAINTENANCE-DECISION CONSEQUENCES

We call the following a *terminal-instant decision stress test*: one decision per test engine, at the last available window, under a fixed rule. It measures wrong triggers at a given state; it is not a simulation of first maintenance time, fleet availability

TABLE VI
DRIFT CONTROLS AT 5% OF RANGE, FD002, HETEROSCEDASTIC HEAD.
PICP UNDER EACH PERTURBATION.

Perturbation	f_{oob}	LSTM	Transformer
Ramp (per window)	0.114	0.048	0.025
Ramp (contin. trajectory)	0.141	0.042	0.024
Bias, equal end value	0.150	0.486	0.230
Reversed ramp	0.111	0.679	0.691
Shuffled ramp	0.112	0.285	0.123
Single-channel ramp (s_9)	7.4e-05	0.629	0.538

or lifetime cost. The rule triggers maintenance when a one-sided 95% lower bound on RUL is at or below the lead time L . For the Gaussian mechanisms the bound is $\hat{\mu} - 1.645\hat{\sigma}$; for split conformal it is a one-sided conformal quantile at $\alpha = 0.05$ recalibrated for the purpose, so that every mechanism is compared at the same nominal tail risk. An engine with true $\text{RUL} \leq L$ that is not triggered is an *unrecognised maintenance need*; we report its rate over all engines and conditionally over the at-risk group. An engine triggered with true $\text{RUL} > L + 20$ is a *premature trigger*, for which we report the true RUL at trigger, both under the 125-cycle label clip and unclipped, and its excess over L ; the excess is the life surrendered relative to an ideal trigger at exactly L , on the assumption that maintenance follows immediately, not the total remaining life. Costs are $c_{\text{late}}r_{\text{late}} + c_{\text{early}}r_{\text{early}}$ at cost ratios $\{5, 20, 100\}$, and rankings between mechanisms are compared with a per-engine bootstrap.

Table VII gives the heteroscedastic head; the full 5 mechanisms $\times$ 2 backbones $\times$ 4 sub-datasets $\times$ 4 conditions $\times$ 3 lead times with costs and bootstrap intervals are in the supplementary material. Two results are robust. First, the direction of error depends on the degradation. Under bias the unrecognised rate is non-zero (0.078 overall, 0.37 within the at-risk group, FD002/LSTM at $L = 20$); the deep ensemble has the lowest unrecognised rate in all eight backbone–dataset combinations. Under drift the unrecognised rate is zero in every cell, and instead 52–69% of six-condition engines are triggered at $L = 20$ (40–72% across the three lead times), with a mean unclipped true remaining life among those triggers of 101–105 cycles (94–96 under the label clip) on both backbones; the single-condition sub-datasets are far less affected (2.6% and 14% premature on FD001 and FD003 for the LSTM). Drift makes the models pessimistic, with the learned scale contracted as well; bias makes them miss. Second, no mechanism is recommendable on cost. Across the 96 (backbone, sub-dataset, condition, lead time) combinations the cost ranking reverses between cost ratios in 36, and in 20 of those the top-ranked mechanism changes; the per-engine bootstrap finds only 63–101 of 144 pairwise top-versus-rest differences to exclude zero at 95%. The qualitative asymmetry is the finding; the ranking is not.

The two structured degradations therefore fail on opposite sides of the rule. Bias produces the safety-side error, an engine operated past the point at which service was due. Drift produces an economic error that a coverage statistic does not convey: between half and two thirds of a six-condition

fleet triggered early, surrendering on average about 84 cycles beyond the lead time. Because true degradation also produces trends, a monitor that distinguishes sensor drift from engine degradation is a research problem rather than a corrective we can offer here; what the stress test establishes is that a decision audit must count both sides of the rule, and that neither side is predicted by the excursion statistic.

X. SPLIT CONFORMAL PREDICTION ON THE STANDARD PARTITION

Split conformal prediction guarantees marginal coverage of at least $1 - \alpha$ when calibration and test scores are exchangeable [25]. On the standard C-MAPSS partition the calibration windows are drawn across whole trajectories and the test windows are concentrated near truncation points, and windows within an engine are dependent, so the conditions of the guarantee are not obviously met. We observe empirical coverage of 0.932 (CP-norm, LSTM, FD002; +0.032); 0.866 (CP-norm, Transformer, FD004; −0.034); the other 6 backbone–dataset combinations lie within ± 0.03 . We do not read these deviations as a failure of the guarantee: over-coverage is permitted by it, and a single finite-sample empirical coverage does not establish non-exchangeability in either direction. What the observations do establish is that a conformal interval reported on this partition should be accompanied by its empirical coverage, with a per-engine-clustered uncertainty, rather than by the nominal level alone.

XI. DISCUSSION

What a reliability engineer should change: Three practices follow from the results. First, validate interval mechanisms under fair calibration and score them with a proper interval score; a coverage figure above nominal is not evidence of a better interval, and a mechanism that draws its scale from training residuals is not comparable with one that uses held-out data. Second, do not read a range-based excursion statistic as a risk indicator. On single-condition data some damaging drifts leave it at zero; on six-condition data bias and drift produce comparable excursion with opposite decision consequences. A monitor that separates sensor drift from genuine degradation, both of which trend, is needed and is not provided here. Third, expect drift and bias to fail on opposite sides of the decision rule, and audit both premature triggers and unrecognised needs rather than either alone; do not select a mechanism on a single cost ratio.

Protocol: The controlled 2×2 experiment shows that of the two choices only the checkpoint criterion matters materially, that its effect is far larger on the asymmetric score than on RMSE, and that it degrades calibration as a side effect. A published C-MAPSS result cannot be corrected for it after the fact; the protocol must be stated.

On the learned scale as a measurement uncertainty: A heteroscedastic $\hat{\sigma}$ is tempting to read as a GUM combined standard uncertainty. The results argue against it: the model’s own uncertainty, as evidenced by a protocol effect of up to 14% on the same architecture, is not negligible in the sense the GUM presumes [32], and a scale that contracts under

TABLE VII

TERMINAL-INSTANT DECISION STRESS TEST AT LEAD TIME $L = 20$, HETEROSCEDASTIC HEAD, ONE-SIDED 95% LOWER BOUND. UNRECOGNISED: FRACTION OF ALL ENGINES (IN PARENTHESES, OF ENGINES WITH TRUE RUL $\leq L$) NOT TRIGGERED. PREMATURE: FRACTION TRIGGERED WITH TRUE RUL $> L + 20$. TRUE RUL AT TRIGGER IS GIVEN BOTH WITH THE 125-CYCLE LABEL CLIP USED ELSEWHERE IN THE PAPER AND WITH THE ORIGINAL UNCLIPPED LABEL, AS THE MEAN OVER PREMATURE TRIGGERS; THE EXCESS OVER L USES THE UNCLIPPED VALUE. TRIGGER DECISIONS USE THRESHOLDS OF AT MOST 50 CYCLES AND ARE UNAFFECTED BY THE CLIP.

Condition	Data/backbone	Unrecognised (cond.)	Premature	RUL at trigger (clipped / unclipped)	Excess over L (unclipped)
Clean	FD002/LSTM	0.000 (0.000)	0.037	49.4 / 49.4	29.4
Bias 5%	FD002/LSTM	0.078 (0.367)	0.170	94.3 / 103.7	83.7
Drift 5%	FD002/LSTM	0.000 (0.000)	0.693	95.1 / 104.8	84.8
Drift 5%	FD004/LSTM	0.000 (0.000)	0.590	96.0 / 104.7	84.7
Drift 5%	FD002/Transformer	0.000 (0.000)	0.692	95.0 / 104.6	84.6
Drift 5%	FD001/LSTM	0.000 (0.000)	0.026	62.0 / 62.0	42.0
Drift 5%	FD003/LSTM	0.000 (0.000)	0.138	55.1 / 55.1	35.1

drift violates the requirement that a stated uncertainty not be optimistic where the measurement is least trustworthy. On clean, in-range inputs $1.645 \hat{\sigma}$ is an empirically calibrated half-width of a central 90% interval, and nothing more; the ten-point loss thresholds of Section VII are properties of this benchmark, not a general range of validity.

XII. LIMITATIONS

Two backbones on simulated data; N-CMAPSS [34] with recorded flight profiles is the next step. Because each seed draws its own partition, the five-seed spread mixes initialisation and partition variability; Table III shows that on FD001 single models on the seed-42 partition average 60.2 IS against 67.7 over five partitions, a difference comparable with any mechanism margin. Five seeds also do not resolve that spread: a five-seed standard deviation is itself uncertain by about $\pm 35\%$ (Appendix D). The ensemble's main-table entries are single point estimates, with between-ensemble spread of up to two points of PICIP from disjoint groups of a fifteen-seed pool. Structured degradation was applied to 5% of the training range, which is not an instrument's full scale; the excursion at which coverage drops ten points is interpolated and becomes unreliable where curves are flat, so values beyond 10% are reported as bounds. The drift controls were run on two sub-datasets. The maintenance rule is one fixed rule with one lead-time family; rank reversal across cost ratios was checked exhaustively over the 96 combinations. KernelSHAP attribution was computed and is not reported: cross-seed rank correlation was about 0.3 and did not improve with a fourfold sampling budget. No stable latency plateau was found for the Transformer at any batch size tested. Every perturbation is seeded from a SHA-256 digest of its (sub-dataset, condition, trial) identifiers, dropout sampling is seeded, and cuDNN is run in deterministic mode; two independent runs of the released code on the same machine (PyTorch 2.11, CUDA 13, RTX A5000 Laptop GPU) reproduced 71 of the 74 result files behind the tables bit-for-bit and the remainder in every field except a wall-clock timing field, as listed with their hashes in REPRODUCE.md. Training, timing and cross-platform reproduction are outside that check. Each noisy condition is nevertheless one seeded realisation of five trials; cross-trial standard deviations are stored with every result, and threshold counts and attribution shares move by a few cells or points between independent seedings.

XIII. CONCLUSION

Under a leakage-audited protocol with fair calibration, which mechanism gives the best-covered RUL interval depends on the backbone; by interval score the deep ensemble is best in every combination in sample, with a margin that a paired comparison against independently seeded ensembles favours in point estimate on all four sub-datasets but establishes on none. Coverage lost under additive noise is lost predominantly through the mean, robustly to decomposition order, in all 16 cases by point estimate and in 32 of 32 order-specific cells by paired interval. Different sensor degradations produce different directions of prediction and decision error: a within-window drift, damaging through its temporal structure, makes the models pessimistic and triggers premature maintenance on half to two thirds of a six-condition fleet, while bias makes them miss maintenance needs; on single-condition data drift does this at an input excursion below 10^{-4} . A single excursion statistic does not characterise these risks, cost rankings between mechanisms are unstable across cost ratios, and monitoring that separates sensor drift from engine degradation remains to be developed.

APPENDIX A AUDIT HISTORY

The protocol of Section III was adopted after an audit of the pipeline behind an earlier, unpublished version of this work found that the checkpoint had been selected on the official test set and the normaliser fitted on data used for checkpoint selection; the controlled 2×2 experiment of Section III-C reproduces those two choices, one at a time, with the fitting engines held fixed. The same audit found that the earlier MC Dropout baseline had omitted the aleatory variance term, producing coverage of 0.40–0.49, and that an earlier ensemble evaluation had given each member its own noise realisation, which averages the perturbation at $1/\sqrt{M}$ and manufactured an apparent robustness. An earlier mean-only counterfactual used a separately trained MSE model and disagreed in sign with the frozen-output control on two of three sub-datasets. All are corrected above; the earlier results are retained in the released repository (<https://github.com/jeff-1980/rul-interval-reliability>, tag v1.5-submission) under results/superseded/.

APPENDIX B

FAIR CALIBRATION VERSUS TRAINING RESIDUALS

Table VIII gives the ratio of the calibration-set to the training-residual scale σ_{res} , and the resulting PICP/MPIW under each estimator, for the fixed-variance and MC Dropout mechanisms.

The training-residual PICP is below the calibration-set PICP in all sixteen (backbone, dataset, mechanism) cells, confirming systematic under-coverage of the training-residual estimator. MPIW under the training-residual estimator is narrower throughout (e.g. LSTM/FD001 fixed variance: 38.55 versus 47.82 cycles), consistent with the narrower scale.

APPENDIX C

TRANSFORMER ATTRIBUTION

Table IX gives the four-way decomposition for the Transformer, in the same form as Table IV for the LSTM. The mean effect dominates under both decomposition orders in all eight (sub-dataset, mechanism) cells, with no exception.

APPENDIX D

SEED VARIANCE AND ENSEMBLE SIZE

Ten further seeds per sub-dataset were trained under the same protocol (nine of 120 QC checks fell outside the five-seed range and were retained). The ratio of fifteen- to five-seed PICP standard deviation is 1.36, 0.70, 0.80 on FD001, FD002 and FD004, consistent with the sampling error of a standard deviation from $n = 5$. Over $M \in \{2, 3, 5, 10\}$ ensemble size yields diminishing but monotone gains on every sub-dataset; on FD004, PICP rises from 0.909 to 0.924 and IS falls from 77.6 to 75.0. Three disjoint five-member ensembles from the pool give IS of 56.7/55.1/60.3, 60.4/58.7/56.8, 52.2/54.2/55.8, 75.2/74.8/77.4 on FD001–FD004.

APPENDIX E

LATENCY

Per-sample latency of every mechanism was measured on one NVIDIA RTX A5000 Laptop GPU at batch sizes 1, 32 and 512 with CUDA-event and wall-clock timing separately, eleven repetitions per cell, reported as median and interquartile range without exclusion. Table X gives the single-pass (heteroscedastic head) CUDA-event median and interquartile range, in microseconds per sample, at each batch size; wall-clock medians agree with CUDA-event medians to within a few percent throughout and are not tabulated separately. The architectural multipliers $T = 50$ (MC Dropout) and $M = 5$ (ensemble) are consistent with the LSTM measurements and are used as a diagnostic only, not as an acceptance criterion.

The LSTM's single pass costs 0.0018 ms per sample on FD001 and 0.0028 ms on the other sub-datasets at batch 512 and about 0.35–0.55 ms at batch 1; the Transformer's 2.5–3 times more. One cell (Transformer, FD002, batch 512) returned a median 4–5 times its neighbours and is reported as measured.

REFERENCES

- [1] A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage propagation modeling for aircraft engine run-to-failure simulation," in *Proc. 1st Int. Conf. PHM*, pp. 1–9, 2008.
- [2] Z. Li, S. Luo, H. Liu, C. Tang, and J. Miao, "TTSNet: Transformer-temporal convolutional network-self-attention with feature fusion for prediction of remaining useful life of aircraft engines," *Sensors*, vol. 25, p. 432, 2025. <https://doi.org/10.3390/s25020432>.
- [3] X. Chen, "A novel transformer-based DL model enhanced by position-sensitive attention and gated hierarchical LSTM for aero-engine RUL prediction," *Sci. Rep.*, vol. 14, p. 10061, 2024. <https://doi.org/10.1038/s41598-024-59095-3>.
- [4] S. Deng and J. Zhou, "Prediction of remaining useful life of aero-engines based on CNN-LSTM-Attention," *Int. J. Comput. Intell. Syst.*, vol. 17, p. 232, 2024. <https://doi.org/10.1007/s44196-024-00639-w>.
- [5] M. Ruvaifa *et al.*, "Evaluating the impact of data preprocessing on deep neural architectures for remaining useful life estimation of turbofan engines," *Array*, 2026. Article 100919.
- [6] L. Basora, A. Viens, M. A. Chao, and X. Olive, "A benchmark on uncertainty quantification for deep learning prognostics," *Reliab. Eng. Syst. Saf.*, vol. 253, p. 110513, 2025.
- [7] Y. Zhan, Z. Kong, Z. Wang, X. Jin, and Z. Xu, "Remaining useful life prediction with uncertainty quantification based on multi-distribution fusion structure," *Reliab. Eng. Syst. Saf.*, vol. 251, p. 110383, 2024.
- [8] W. Peng, Z. S. Ye, and N. Chen, "Bayesian deep-learning-based health prognostics toward prognostics uncertainty," *IEEE Trans. Ind. Electron.*, vol. 67, pp. 2283–2293, 2019.
- [9] S. B. Ayed, R. Sadeghian Broujeny, and R. T. Hamza, "Remaining useful life prediction with uncertainty quantification using evidential deep learning," *J. Artif. Intell. Soft Comput. Res.*, vol. 15, pp. 37–55, 2025.
- [10] S. Piao, R. Huang, and F. Tsung, "CRULP: Reliable RUL estimation inspired by conformal prediction," *IEEE Trans. Instrum. Meas.*, vol. 74, pp. 1–11, 2025.
- [11] S. Mao, X. Li, and B. Zhao, "Remaining useful life prediction based on time-series features and conformalized quantile regression," *Meas. Sci. Technol.*, vol. 35, no. 12, p. 126113, 2024.
- [12] M. Shamim *et al.*, "Leakage-robust evaluation and data-scale sensitivity of attention-enhanced multi-task learning for joint fault diagnosis and remaining useful life estimation," arXiv:2607.16493, 2026.
- [13] H. Hachem, H. C. Vu, and M. Fouladirad, "Different methods for RUL prediction considering sensor degradation," *Reliab. Eng. Syst. Saf.*, vol. 243, p. 109897, 2024.
- [14] S.-S. Chen and X.-Y. Li, "Modeling dependent degradation data considering inherent causal relationships for reliability analysis and remaining useful life prediction," *Reliab. Eng. Syst. Saf.*, vol. 277, p. 113080, 2027. DOI registered June 2026; print issue January 2027.
- [15] X. Li, Q. Ding, and J. Q. Sun, "Remaining useful life estimation in prognostics using deep convolution neural networks," *Reliab. Eng. Syst. Saf.*, vol. 172, pp. 1–11, 2018.
- [16] H. Li, W. Zhao, Y. Zhang, and E. Zio, "Remaining useful life prediction using multi-scale deep convolutional neural network," *Appl. Soft Comput.*, vol. 89, p. 106113, 2020.
- [17] J. Zhang, P. Wang, R. Yan, and R. X. Gao, "Long short-term memory for machine remaining life prediction," *J. Manuf. Syst.*, vol. 48, pp. 78–86, 2019.
- [18] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, and J. Lin, "Machinery health prognostics: a systematic review from data acquisition to RUL prediction," *Mech. Syst. Signal Process.*, vol. 104, pp. 799–834, 2018.
- [19] R. Zhao, R. Yan, Z. Chen, K. Mao, P. Wang, and R. X. Gao, "Deep learning and its applications to machine health monitoring," *Mech. Syst. Signal Process.*, vol. 115, pp. 213–237, 2019.
- [20] X. Liu, Z. Zhang, Z. Li, J. Wang, Y. Zhu, and H. Ma, "Advancements in bearing health monitoring and remaining useful life prediction: techniques, challenges, and future directions," *Meas. Sci. Technol.*, vol. 36, p. 032003, 2025.
- [21] Y. Gal and Z. Ghahramani, "Dropout as a bayesian approximation: representing model uncertainty in deep learning," in *Proc. ICML*, vol. 48, pp. 1050–1059, 2016.
- [22] B. Lakshminarayanan, A. Pritzel, and C. Blundell, "Simple and scalable predictive uncertainty estimation using deep ensembles," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.
- [23] D. A. Nix and A. S. Weigend, "Estimating the mean and variance of the target probability distribution," in *Proc. ICNN*, pp. 55–60, 1994.

TABLE VIII
CALIBRATION-SET VERSUS TRAINING-RESIDUAL σ_{res} : RATIO, AND PICP/MPiW UNDER EACH ESTIMATOR. TR.: TRAINING-RESIDUAL; CAL.: CALIBRATION-SET (FAIR, USED IN TABLE II).

Backbone/Data	Ratio	Fixed variance		MC Dropout	
		PICP Tr.	PICP Cal.	PICP Tr.	PICP Cal.
LSTM/FD001	1.241	0.806	0.870	0.830	0.898
LSTM/FD002	1.125	0.880	0.916	0.894	0.926
LSTM/FD003	1.364	0.818	0.882	0.840	0.894
LSTM/FD004	1.103	0.854	0.883	0.865	0.893
Transformer/FD001	1.079	0.870	0.884	0.912	0.920
Transformer/FD002	1.084	0.897	0.918	0.903	0.922
Transformer/FD003	1.238	0.788	0.858	0.838	0.892
Transformer/FD004	1.168	0.814	0.856	0.847	0.888

TABLE IX
FOUR-WAY ATTRIBUTION OF COVERAGE LOSS UNDER ADDITIVE NOISE, TRANSFORMER, AT THE TEN-POINT LOSS OF THE HETEROSCEDASTIC HEAD. SCALE CONTRIBUTION IS THE SHARE OF TOTAL LOSS UNDER EACH DECOMPOSITION ORDER.

		FD001	FD002	FD003	FD004
Heteroscedastic	scale, mean first	+20.0%	+11.4%	+0.2%	+3.2%
	scale, scale first	+20.3%	+31.0%	+21.0%	+21.9%
CP-norm	scale, mean first	+17.7%	+11.6%	+6.5%	+4.3%
	scale, scale first	+24.9%	+32.5%	+16.8%	+23.7%
Mean dominates (point estimate), both orders		yes	yes	yes	yes
CI excludes zero, mean first (het./CP)		yes/yes	yes/yes	yes/yes	yes/yes
CI excludes zero, scale first (het./CP)		yes/yes	yes/yes	yes/yes	yes/yes

TABLE X
PER-SAMPLE LATENCY OF THE HETEROSCEDASTIC HEAD, CUDA-EVENT TIMING, MEDIAN (IQR) IN MICROSECONDS, AT BATCH SIZES 1, 32 AND 512.

Data	Backbone	Batch 1	Batch 32	Batch 512
FD001	LSTM	553.89 (71.65)	10.98 (1.04)	1.80 (0.12)
FD001	Transformer	802.82 (67.09)	21.98 (1.88)	5.22 (0.09)
FD002	LSTM	364.54 (46.51)	12.77 (3.09)	2.80 (0.06)
FD002	Transformer	827.39 (76.24)	25.63 (2.13)	27.94 (1.07)
FD003	LSTM	351.23 (13.31)	12.13 (1.17)	2.81 (0.04)
FD003	Transformer	755.71 (126.98)	25.32 (1.04)	7.06 (0.05)
FD004	LSTM	355.33 (24.03)	11.07 (1.14)	2.82 (0.08)
FD004	Transformer	887.81 (86.02)	25.25 (2.43)	6.15 (0.11)

- [24] A. Kendall and Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?," in *Adv. Neural Inf. Process. Syst.*, vol. 30, 2017.
- [25] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Conformal prediction beyond exchangeability," *Ann. Statist.*, vol. 51, pp. 816–845, 2023.
- [26] J. Gawlikowski *et al.*, "A survey of uncertainty in deep neural networks," *Artif. Intell. Rev.*, vol. 56, pp. 1513–1589, 2023.
- [27] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, and J. Snoek, "Can you trust your model's uncertainty? evaluating predictive uncertainty under dataset shift," in *Adv. Neural Inf. Process. Syst.*, vol. 32, 2019.
- [28] M. Valdenegro-Toro and D. Saromo Mori, "A deeper look into aleatoric and epistemic uncertainty disentanglement," in *Proc. CVPR Workshops*, pp. 1508–1516, 2022.
- [29] M. Seitzer, A. Tavakoli, D. Antic, and G. Martius, "On the pitfalls of heteroscedastic uncertainty estimation with probabilistic neural networks," in *Proc. ICLR*, 2022.
- [30] A. Stirn, H.-H. Wessels, M. Schertzer, L. Pereira, N. E. Sanjana, and D. A. Knowles, "Faithful heteroscedastic regression with neural networks," in *Proc. AISTATS*, 2023.
- [31] JCGM 100:2008, "Evaluation of measurement data — Guide to the Expression of Uncertainty in Measurement (GUM)," tech. rep., Joint Committee for Guides in Metrology, 2008.
- [32] L. Angrisani *et al.*, "Addressing nonnegligible model uncertainty in machine learning-assisted measurements," *IEEE Trans. Instrum. Meas.*, 2026.
- [33] T. Gneiting and A. E. Raftery, "Strictly proper scoring rules, prediction, and estimation," *J. Am. Stat. Assoc.*, vol. 102, pp. 359–378, 2007.
- [34] M. A. Chao, C. Kulkarni, K. Goebel, and O. Fink, "Aircraft engine run-to-failure dataset under real flight conditions for prognostics and diagnostics," *Data*, vol. 6, p. 5, 2021.